

MIR SAYEED MOHAMMAD

Graduate Research Assistant, Ph.D. Student (ECE) | Georgia Tech, Atlanta, Georgia, USA

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I design **real-time**, **energy-efficient** LiDAR/radar/camera perception on **embedded** devices; specialize in **multi-sensor fusion**, **spatiotemporal** models, and **edge** deployment for **autonomous vehicles** and robots.

PROFESSIONAL EXPERIENCE

- Aug 2023-Present** **Graduate Research Assistant** | ECE, Georgia Tech | **Advisor** - Dr. Saibal Mukhopadhyay.
- ▷ Designed **SSMRadNet** for real-time radar detection | ↓ **88×** compute ↑ **3.7×** fps than transformers.
 - ▷ Developed **multimodal** sensor suite with **radar+LiDAR+stereo** camera for robot perception.
 - ▷ Improved efficiency of **adaptive LiDAR 3D object detection** by **40%** with $\Delta AP \leq 2\%$.
 - ▷ Deployed adaptive LiDAR detection on **Boston Dynamics Spot** robot for **autonomous** navigation.
- Jun 2022-Jul 2023** **Lecturer** | Computer Science and Engineering, BRAC University, Dhaka, Bangladesh.
- ▷ Taught digital electronics, digital logic and microprocessor lab.

SELECTED PROJECTS

- Aug 2024-Present** **Information-Aware Multi-Sensor Fusion** – Designing LiDAR–Radar–Camera fusion with **sensor-specific vs. sensor-agnostic** token disentanglement; improving weather, noise and **contention** robustness; reducing memory usage, computation and latency of fused pipeline.
- Aug 2024-Present** **State-Space Modeling of Asynchronous Perception** – Deploying edge hardware-friendly **Mamba/SSM** temporal encoders for 3D detection+segmentation replacing 3D-CNN/transformer blocks; increasing overall system throughput in **multi-rate** and **multi-spectral** radar, LiDAR, event camera and infrared perception systems.
- Aug 2023-Jul 2024** **Sensor Programming & Collaborative Control** – Built closed-loop LiDAR perception–control on **Boston Dynamics Spot & LoCoBot** with **ROS** and CUDA-accelerated computation; reduced CPU/GPU utilization by **40%** while maintaining **10 Hz** control rate.
- May 2021-Apr 2022** **Autonomous Driving via Curriculum, Ensemble & Multitask RL** (*B.Sc. Thesis*) – Trained urban-driving **PPO** policy with curriculum and **Mixture-of-Experts**; achieved **84%** success across traffic/scene variations with reduced infraction rate; multi-task heads for lane-keeping, collision and traffic avoidance.

PUBLICATIONS

- Sep 2025** SSMRadNet: A Sample-wise State-Space Framework for Efficient and Ultra-Light Radar Segmentation and Object Detection. | [WACV-2026](#)
- May 2025** Adaptive-Cloud: Dynamic Computation Control for 3D Object Detection from LiDAR Point Clouds. | [IEEE Robotics and Automation Letters](#)
- Jun 2023** A Sequence Agnostic Multimodal Preprocessing for Clogged Blood Vessel Detection in Alzheimer's Diagnosis. | [IEEE ICASSPW - 2023](#)
- Nov 2022** S2C-DeLeNet: A Segmentation-Classification Integration for Skin Cancer Detection from Dermoscopic Images. | [Computers in Biology and Medicine — Elsevier](#)

EDUCATION

- Aug 2023-Present** **Ph.D. student** | Electrical and Computer Engineering, Georgia Tech.
Coursework- Robotics, Mechatronics, Networked Control, Generative & Geometric Deep Learning.
- May 2025** **M.Sc.** | ECE, Georgia Tech. **CGPA: 4.00/4.00**
- May 2023** **B.Sc.** | EEE, Bangladesh Univ. of Engineering and Technology. **CGPA: 3.97/4.00** (Top 5%)

TECHNICAL SKILLS

Programming and DL. - C, C++, Java, Python, PyTorch, Tensorflow, CNN, RNN, Transformers, SSM, RL.
Design and Sim. - MATLAB, Cadence, Multisim, Proteus, Autodesk Eagle, Quartus, Solidworks, AutoCAD
Robotics Frameworks - ROS, NVIDIA Isaacsim, Gazebo, CoppeliaSim, Webots
Embedded Platforms - ATmega, STM32, ESP8266/32, RaspberryPi, Jetson, TI AWR1843/DCA1000.